

ZHAW SCHOOL OF ENGINEERING
INSTITUTE OF MECHATRONIC SYSTEMS

Betaflight – Offline Controller Tuning

Data-Driven Tuning of Rate and Altitude Controllers for FPV Drones

Bachelor Thesis

Yuri Bianchi, Dario Jurietti and Janick Dort

Supervisor: Michael Peter (pmic)

Co-Supervisor: Prof. Dr. Ruprecht Altenburger (altb)

March 31, 2026

Declaration of Originality

We hereby declare that this thesis is our own work and has been composed solely by the undersigned. We have not used any sources or aids other than those indicated. All passages taken verbatim or in substance from published or unpublished works of others have been clearly marked as such. This thesis has not been submitted, in whole or in part, for any other degree or examination at this or any other institution.

Winterthur, March 31, 2026

Yuri Bianchi

Dario Jurietti

Janick Dort

Zusammenfassung

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Abstract

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

Contents

Declaration of Originality	ii
Zusammenfassung	iii
Abstract	iv
List of Figures	vi
List of Tables	vii
Preface	viii
1 Introduction	1
1.1 Initial situation	2
1.2 Task and Objective	2
2 Theoretical Foundations	3
2.1 Chirp Signal	3
2.1.1 Mathematical Definition	4
2.1.2 Common Frequency Profiles	5
2.1.3 Lag Filter for Chirp Input Shaping	7
2.2 Signal Processing	8
2.2.1 Baseband Filtering via Frequency Rotation	9
2.2.2 Frequency Response Estimation	11
2.2.3 Spectral Estimation using Welch's Method	12
2.3 Control Loop Calculations	14
2.3.1 Calculation of the Control Loop Components	14
2.3.2 Closed-Loop Performance Metrics	15
3 Methodology	17
3.1 Angle Mode	17
3.1.1 Difference between Angle and Acro Mode	17
3.1.2 Control Loops in Angle Mode	17
3.1.3 Derivation of the Target Angle	18
3.1.4 Chirp Signal for System Identification	19
3.1.5 Measurement of the Control Loop Components	20
3.1.6 Theoretical Expectation	21
Declaration of Generative AI Use	22

List of Figures

2.1	Linear chirp signal used for the simulation.	5
2.2	Exponential chirp signal used for the simulation.	6
2.3	Exponential chirp signal after conditioning with the lag filter used for chirp input shaping.	7
2.4	Simulation model used for the signal processing analysis.	8
2.5	Time-domain signal before (top) and after (bottom) baseband filtering. The filtering reduces noise while preserving the main signal characteristics.	10
2.6	Bode plot of the true and estimated transfer function using Welch's method. The agreement is improved due to the reduced influence of noise.	13
2.7	Magnitude-squared coherence between input and output signals. High values indicate a reliable transfer function estimate.	14
3.1	Cascaded control structure of the Betaflight Angle mode controller.	18
3.2	Current control loop structure in Betaflight when the chirp signal is injected inside the closed loop.	19

List of Tables

Preface

This thesis was carried out during the spring semester 2026 at the Institute of Mechatronic Systems (IMS) at the ZHAW School of Engineering. It continues the work started in the preceding project thesis conducted in the fall semester 2025.

We would like to thank our supervisor, Michael Peter, for...

1 Introduction

First-Person-View (FPV) racing drones require precise and responsive control to achieve stable flight behaviour, suppress disturbances such as propeller wash, and maintain agility during rapid manoeuvres. The flight performance strongly depends on the tuning of the flight controller's feedback loops. In particular, the parameters of the angular rate controller determine how quickly and accurately the drone reacts to pilot inputs and external disturbances.

In practice, controller tuning is still often performed manually through iterative trial-and-error procedures. This requires extensive flight testing and considerable piloting experience, making the process time-consuming and difficult to reproduce. Although modern flight controller firmware such as Betaflight provides extensive configuration options and detailed flight data logging, systematic tuning remains a challenge.

One commonly used tool for offline tuning is the PID-Toolbox. In this approach, flight logs are recorded during normal flights and later analysed to derive suitable controller parameters. However, the system excitation depends entirely on the pilot's inputs and the natural flight behaviour of the drone. As a result, not all relevant frequency ranges are sufficiently excited, which can limit the quality of the measurements.

In the approach proposed in this thesis, a chirp excitation signal is injected directly into the control loop. This allows the system to be excited over a wide range of frequencies in a controlled and systematic way, resulting in higher-quality measurements for system identification and controller tuning.

A previous project introduced a proof-of-concept approach for offline tuning of angular rate controllers based on recorded flight data. In the preceding project thesis, this concept was restructured and modularised and now serves as the foundation for the present work.

The objective of this bachelor thesis is to further refine and extend this approach into an offline tuning framework for multiple control loops used in Betaflight-based flight controllers. In addition to the existing angular rate controller tuner, the framework will be extended with code modules for tuning the angle controller, the altitude hold controller, and the position controller. This extends the tuning framework beyond the inner rate control loop and enables the analysis and tuning of higher-level control layers.

The framework will be implemented primarily in MATLAB and Python and will provide tools for analysing flight logs, estimating system dynamics, and generating controller parameters without requiring repeated flight tests. The software will be developed as an open-source library with accompanying documentation for the FPV community.

By providing a structured, data-driven approach to controller tuning, this project aims to simplify the tuning process and improve flight performance.

1.1 Initial situation

1.2 Task and Objective

2 Theoretical Foundations

In this chapter, the theoretical background of the approach used for the estimation and calculation of the different control loops is described. Since the fundamental approach of all the control loops is the same, the general approach is described here and the specific details of each control loop are described in the Methodology chapter.

To support the theoretical concepts, a simplified simulation model is added. The simulation is gradually developed throughout this chapter, allowing the reader to follow the modelling process step by step and to better understand the behaviour of the control system. This approach should provide a clear and practical understanding of the control loops and the signal processing techniques used in the estimation and calculation of the control loops.

The detailed application of these principles to the different control loops implemented in Betaflight is discussed in the following chapter.

2.1 Chirp Signal

A chirp signal is a signal in which the frequency increases or decreases over time. It is commonly used in system identification and control engineering to analyze the response of a system over a range of frequencies.

The main advantage of a chirp signal is that it allows a wide frequency range to be excited within a single experiment. Instead of testing individual frequencies one by one, the system is continuously stimulated with increasing frequency. This makes the approach both efficient and well suited for practical applications.

By observing how the system responds to this input, it becomes possible to analyse its frequency-dependent behaviour. In particular, the measured input and output signals can be used to estimate the frequency response and, consequently, the transfer function of the system. This provides valuable insight into the system dynamics and forms the basis for the methods described in the following sections.

2.1.1 Mathematical Definition

The instantaneous value of the chirp signal is defined as

$$x(t) = A \cdot \sin(\varphi(t)), \quad (2.1)$$

where A is the amplitude. The amplitude determines the excitation strength of the signal and therefore influences the signal-to-noise ratio of the measurement.

The phase is defined as

$$\varphi(t) = 2\pi \int_0^t f(\tau) d\tau, \quad (2.2)$$

and represents the accumulated instantaneous frequency over time, where $f(\tau)$ denotes the instantaneous frequency at time τ . The specific evolution of the frequency over time will be introduced in the following sections.

In practical implementations such as **Betaflight**, the phase is wrapped once it reaches 2π . This keeps the numerical values bounded and results in a sawtooth-shaped wrapped phase signal, which serves as the argument of the sinusoidal excitation.

The behaviour of the chirp signal is determined by several parameters defining the frequency sweep.

For the simulation model introduced in this chapter, the following parameter values are used:

- f_0 — starting frequency of the chirp signal (simulation: 1 Hz)
- f_1 — final frequency of the chirp signal (simulation: 600 Hz)
- T — total duration of the frequency sweep (simulation: 20 s)
- f_s — sampling frequency used in the discrete-time simulation (simulation: 2000 Hz)
- A — amplitude of the chirp signal (simulation: 230)

These parameters are chosen to reflect the typical operating range of the angular rate and the attitude while ensuring that the relevant frequency dynamics are sufficiently excited.

These parameters directly influence the quality of the system excitation and therefore the accuracy of the subsequent frequency response estimation.

2.1.2 Common Frequency Profiles

The chirp signal can be generated using different frequency profiles, which define how the frequency changes over time. The two most common profiles are the linear chirp and the exponential chirp.

Linear chirp:

For a linear chirp, the instantaneous frequency increases linearly over time.

The instantaneous frequency is therefore given by

$$f(t) = f_0 + \frac{f_1 - f_0}{T} \cdot t, \quad (2.3)$$

and the corresponding phase follows by integrating the frequency over time.

$$\varphi(t) = 2\pi \left(f_0 t + \frac{1}{2} k t^2 \right). \quad (2.4)$$

Figure 2.1 illustrates the resulting chirp signal, the instantaneous frequency $f(t)$, and the corresponding phase trajectory $\varphi(t)$ for the simulation parameters introduced above.

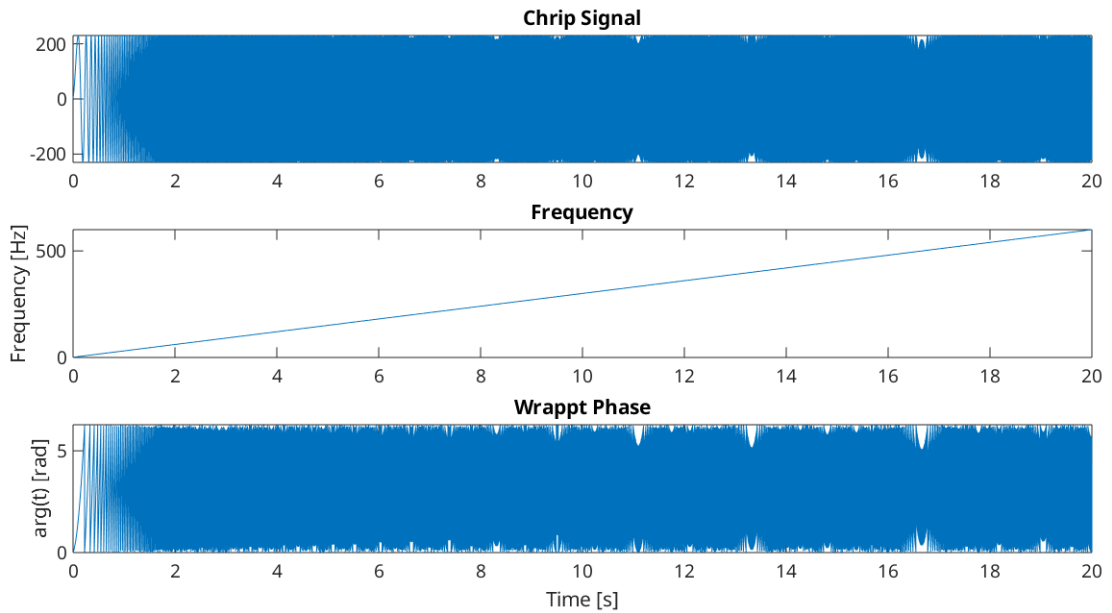


Figure 2.1: Linear chirp signal used for the simulation.

The apparent gaps visible in the chirp signal and in the wrapped phase are not actual discontinuities. They result from the finite sampling time of the simulation and the limited resolution of the plotted data.

Exponential chirp (as used in Betaflight):

For an exponential chirp, the instantaneous frequency increases exponentially from the starting frequency f_0 to the final frequency f_1 over the time interval T . This means that lower frequencies are covered more slowly, while higher frequencies are reached more rapidly.

The instantaneous frequency is defined as

$$f(t) = f_0 \left(\frac{f_1}{f_0} \right)^{t/T}. \quad (2.5)$$

Compared to the linear chirp, this results in a more uniform excitation in logarithmic frequency space, which is particularly useful when analysing systems using Bode plots.

The corresponding phase trajectory is obtained by integrating the frequency

$$\varphi(t) = \frac{2\pi T f_0}{\ln(f_1/f_0)} \left[\left(\frac{f_1}{f_0} \right)^{t/T} - 1 \right]. \quad (2.6)$$

These expressions define the phase trajectory $\varphi(t)$ used both in the simulations and in the time-dependent frequency shifting approach.

The resulting exponential chirp signal, the instantaneous frequency, and the phase trajectory are illustrated in Figure 2.2.

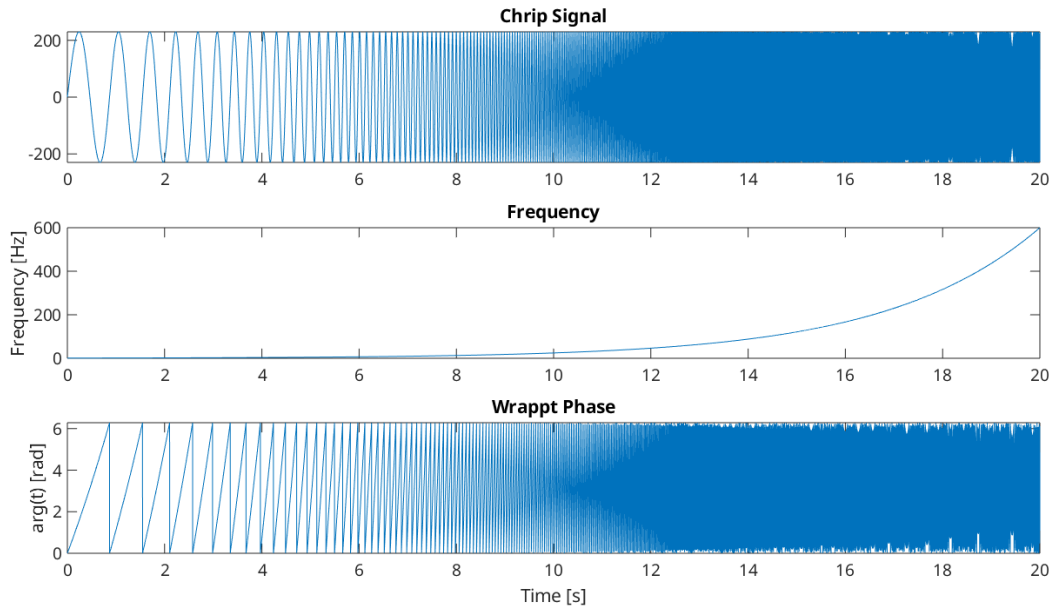


Figure 2.2: Exponential chirp signal used for the simulation.

As with the linear chirp, the apparent gaps in the signal and the wrapped phase are caused by the discrete sampling time and the limited resolution of the plot.

2.1.3 Lag Filter for Chirp Input Shaping

Before the chirp signal is injected into the control loop, it is conditioned by a filter to address the differentiating behavior of the controller effort at low frequencies. This differentiating characteristic causes rising gain at low frequencies, which would lead to excessive motor response at the start of the chirp sweep and potentially cause saturation, compromising the quality of system identification data. The filter reduces low-to-mid frequency components of the chirp signal before injection, such that the combined response produces approximately uniform excitation across the frequency range.

The filter is configured with a pole at 3 Hz and a zero at 30 Hz. This configuration corresponds to the transfer function

$$H(s) = \frac{1 + s/\omega_z}{1 + s/\omega_p} \quad (2.7)$$

where $\omega_z = 2\pi \cdot 30$ rad/s and $\omega_p = 2\pi \cdot 3$ rad/s represent the zero and pole frequencies respectively. The pole introduces the lag component by creating magnitude reduction at frequencies above its cutoff frequency. The zero limits this reduction by introducing magnitude amplification at frequencies above its cutoff frequency. However, since the pole frequency is lower than the zero frequency, the pole's contribution dominates throughout the primary control bandwidth, resulting in a net lag filter. It is important to note that in the Betaflight implementation, the pole frequency (ω_p) must not be set higher than the zero frequency (ω_z), as this would create a lead filter that amplifies high frequencies and could potentially damage the motors.

For the simulation shown in Figure 2.3, the same parameter values introduced in the previous section were used ($f_0 = 1$ Hz, $f_1 = 600$ Hz, $T = 20$ s, $f_s = 2000$ Hz). The excitation signal is generated using the exponential chirp formulation described earlier.

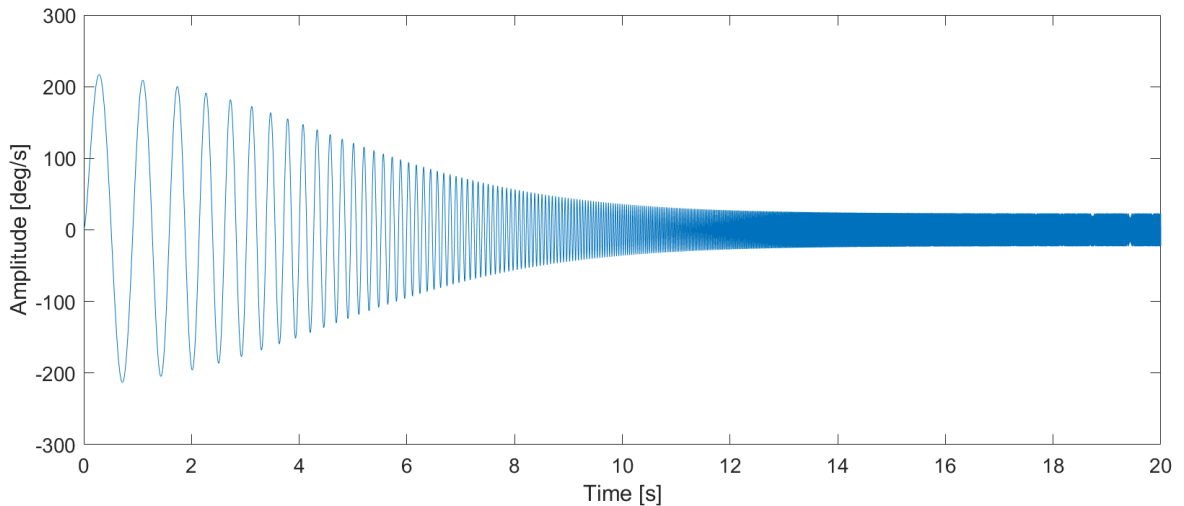


Figure 2.3: Exponential chirp signal after conditioning with the lag filter used for chirp input shaping.

2.2 Signal Processing

To illustrate the influence of the applied signal-processing steps, a simulation is used as a reference. The simulated system represents a simplified model of the plant of the drone's angular-rate controller. It consists of an integrator, a low-pass filter, and a dead-time component. These elements capture the main dynamics of the system and are commonly used in control engineering.

The overall transfer function of the system is given by

$$G(s) = \frac{\omega_1}{s} \cdot \frac{1}{1 + s/\omega_2} \cdot e^{-s/\omega_t}. \quad (2.8)$$

The integrator describes how differences in rotor thrust affect the angular rate over time. The first-order term represents the low-pass behavior introduced by the electronics and actuator dynamics. The dead-time component accounts for delays in the control loop.

For the simulation, the parameters are chosen as

$$\omega_1 = 16 \cdot 2\pi, \quad \omega_2 = 13 \cdot 2\pi, \quad \omega_t = 60 \cdot 2\pi. \quad (2.9)$$

The structure of the simulation model is illustrated in Figure 2.4.

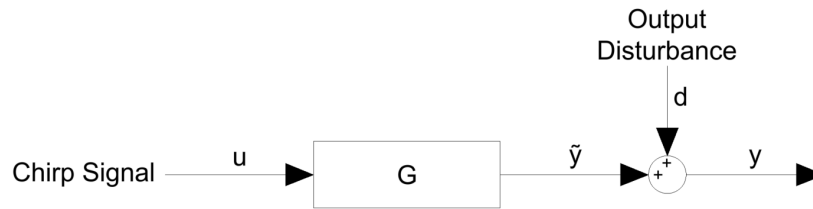


Figure 2.4: Simulation model used for the signal processing analysis.

The chirp signal $u(t)$ is applied to the system, resulting in the ideal output $\tilde{y}(t)$.

To emulate realistic measurement conditions, additive disturbance noise $d(t)$ is superimposed on the output, such that

$$y(t) = \tilde{y}(t) + d(t). \quad (2.10)$$

In the simulation, the disturbance is modelled as additive white Gaussian noise with a standard deviation of $\sigma_d = 2 \text{ deg/s}$. Based on these simulated signals, the following signal-processing steps are applied and evaluated.

2.2.1 Baseband Filtering via Frequency Rotation

Measured signals are often affected by noise and unwanted high-frequency components, which can obscure the relevant information and complicate further analysis. This is also the case in the simulated signals considered here, where additive noise is present. To reduce these disturbances while preserving the phase characteristics of the signal, a frequency-shifting filtering approach is used.

The method is based on shifting the frequency content of the signal to baseband, applying a low-pass filter, and subsequently shifting the signal back to its original frequency range. This allows the filtering operation to focus on a narrow frequency region around the signal of interest.

The underlying principle can be understood using the Fourier shift theorem. Multiplication of a signal $f(t)$ with a complex exponential results in a shift of its spectrum

$$e^{i2\pi\xi_0 t} f(t) \iff \hat{f}(\xi - \xi_0),$$

which shifts the spectrum by the frequency ξ_0 .

In this case, the signal of interest is a chirp signal with a continuously changing instantaneous frequency. Therefore, the frequency shift has to be time-dependent. This is achieved using a complex phasor

$$p(t) = e^{i\phi(t)},$$

where $\phi(t)$ describes the instantaneous phase of the chirp.

Multiplying the measured signal $x(t)$ with this phasor or its complex conjugate results in a rotation of the signal spectrum toward baseband

$$y_R(t) = x(t)p(t), \quad y_Q(t) = x(t)p^*(t).$$

After this rotation, the dominant signal component is centered around 0 Hz, enabling efficient filtering using a low-pass filter.

Once the signal has been shifted to baseband, a low-pass filter is applied to remove high-frequency noise components. Since the useful signal is concentrated near zero frequency, the filter can effectively suppress unwanted spectral components outside this region.

To avoid phase distortion, forward-backward filtering is employed. This approach applies the filter in both forward and reverse directions, resulting in zero-phase filtering.

The filtering operation yields the filtered baseband signals

$$y_R(t) \rightarrow \tilde{y}_R(t), \quad y_Q(t) \rightarrow \tilde{y}_Q(t),$$

which contain the desired signal components with reduced noise.

After filtering, the signal is shifted back to its original frequency range. This is achieved by multiplying the filtered signals with the corresponding inverse phasors and combining the results to obtain a real-valued signal

$$x_f(t) = \text{Re} \left(\frac{1}{2} (\tilde{y}_R(t)p^*(t) + \tilde{y}_Q(t)p(t)) \right).$$

This step reverses the initial frequency rotation and reconstructs a filtered version of the original signal.

The filtered signal is then used for further analysis. By reducing the noise before the spectral estimation, the signal-to-noise ratio is improved, leading to more reliable results.

The effect of the filtering approach is illustrated in Fig. 2.5. The upper plot shows the unfiltered signal, where the noise becomes dominant at later time instances (corresponding to higher chirp frequencies). In contrast, the lower plot shows the filtered signal.

The noise is significantly reduced, especially at higher frequencies, while the main signal characteristics are preserved. The filtered signal closely follows the underlying system response.

At higher frequencies, a slight reduction in amplitude can be observed, which is consistent with the low-pass behavior of the system.

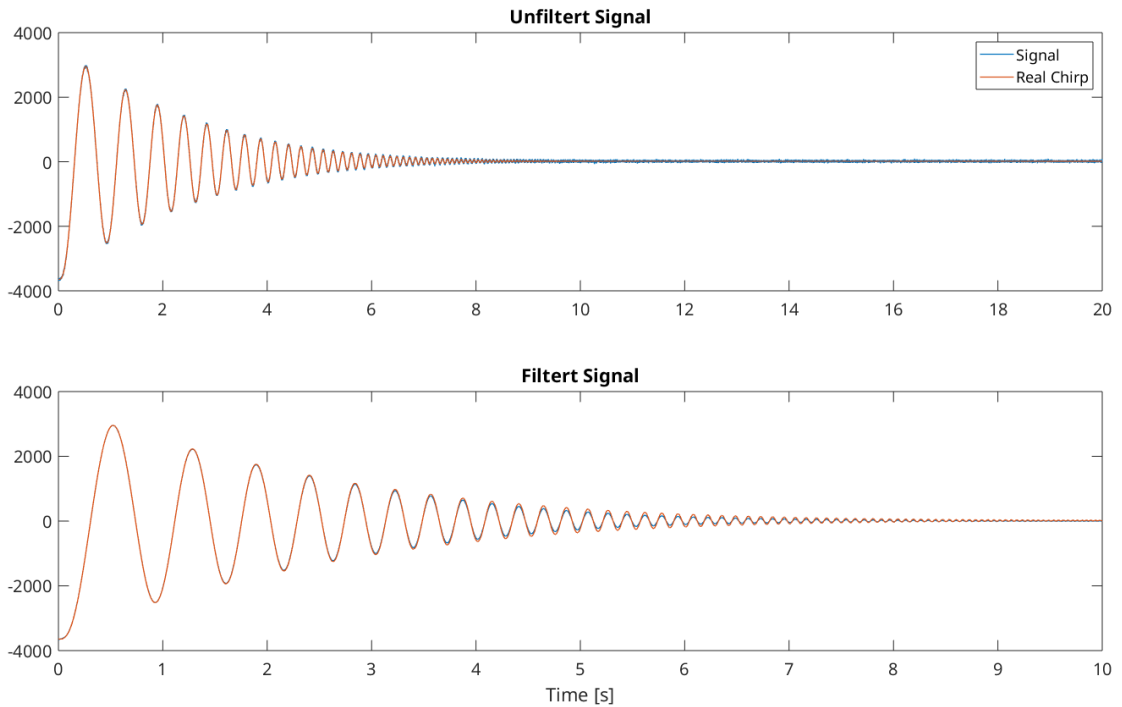


Figure 2.5: Time-domain signal before (top) and after (bottom) baseband filtering. The filtering reduces noise while preserving the main signal characteristics.

2.2.2 Frequency Response Estimation

To estimate the transfer function, the relation between input and output signals is analyzed in the frequency domain. For a linear time-invariant (LTI) system, this relation can be written as

$$G(\omega) = \frac{Y(\omega)}{U(\omega)},$$

where $U(\omega)$ and $Y(\omega)$ denote the Fourier transforms of the input signal $u(t)$ and the output signal $y(t)$, respectively.

In practice, however, the measured output signal is affected by disturbances. Even after applying the baseband filtering, residual noise remains, especially at higher frequencies. The measured output can therefore be written as

$$Y(\omega) = G(\omega)U(\omega) + D_y(\omega),$$

where $D_y(\omega)$ represents an additive disturbance. A direct estimation by dividing $Y(\omega)$ by $U(\omega)$ leads to

$$\frac{Y(\omega)}{U(\omega)} = G(\omega) + \frac{D_y(\omega)}{U(\omega)},$$

which shows that the estimate is directly influenced by noise.

To reduce the sensitivity to noise, spectral quantities can be introduced. However, computing these directly from the Fourier transforms leads to the same issues as the direct division $Y(\omega)/U(\omega)$.

A more robust approach is to estimate the spectral densities using averaging techniques, which reduce the influence of disturbances. This is typically achieved using methods such as Welch's method, which is described in the following section.

In particular, the estimation becomes unreliable in frequency regions where the input signal is small, since the disturbance term can dominate the ratio.

2.2.3 Spectral Estimation using Welch's Method

To obtain reliable estimates of the spectral densities, Welch's method is applied. As discussed in the previous section, direct computation from finite data is sensitive to noise. Therefore, averaging techniques are required to approximate the expected values of the spectral quantities.

The spectral densities describe how the signal energy is distributed over frequency and form the basis for estimating the transfer function.

Let $u[n]$ and $y[n]$ denote the discrete-time input and output signals of length N . In Welch's method, the signals are divided into K overlapping segments of length L . Each segment is multiplied by a window function $w[n]$ to reduce spectral leakage.

The k -th segment of the input signal is given by

$$u_k[n] = u[n + kD] w[n], \quad n = 0, \dots, L - 1, \quad (2.11)$$

where D defines the shift between segments and determines the overlap.

For each segment, the discrete Fourier transform (DFT) is computed as

$$U_k(\omega) = \sum_{n=0}^{L-1} u_k[n] e^{-j\omega n}, \quad Y_k(\omega) = \sum_{n=0}^{L-1} y_k[n] e^{-j\omega n}. \quad (2.12)$$

To obtain a stable representation in the frequency domain, the squared magnitude of the Fourier transform is used. This corresponds to the signal power and makes the spectral estimate less sensitive to noise.

The periodogram estimates are defined as

$$P_{uu,k}(\omega) = \frac{1}{L} |U_k(\omega)|^2, \quad P_{yu,k}(\omega) = \frac{1}{L} Y_k(\omega) U_k^*(\omega) \quad (2.13)$$

Welch's method reduces the variance of the estimates by averaging over all segments. The resulting spectral densities are given by

Welch's method reduces the variance of the estimates by averaging over all segments. The resulting spectral densities are given by

$$S_{uu}(\omega) = \frac{1}{K} \sum_{k=1}^K P_{uu,k}(\omega), \quad S_{yu}(\omega) = \frac{1}{K} \sum_{k=1}^K P_{yu,k}(\omega), \quad (2.14)$$

Using these quantities, the transfer function estimate is computed as

$$G(\omega) = \frac{S_{yu}(\omega)}{S_{uu}(\omega)}. \quad (2.15)$$

The estimated transfer function is shown in Fig. 2.6. Compared to the direct estimation, a significantly better agreement with the true system response can be observed over a wider frequency range.

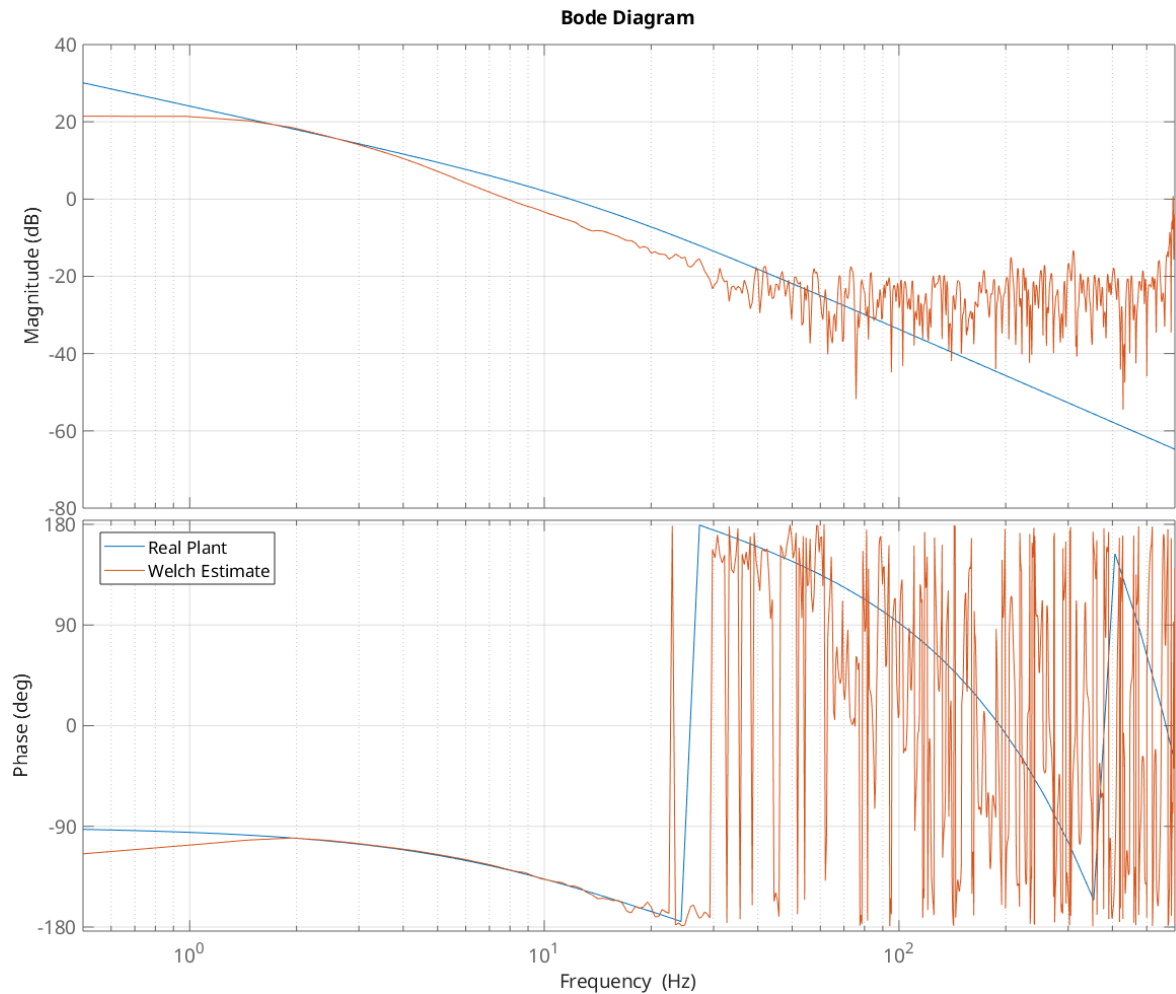


Figure 2.6: Bode plot of the true and estimated transfer function using Welch's method. The agreement is improved due to the reduced influence of noise.

To further assess the quality of the estimate, the coherence function is evaluated. The coherence is defined as

$$\gamma^2(\omega) = \frac{|S_{yu}(\omega)|^2}{S_{uu}(\omega) S_{yy}(\omega)},$$

and takes values in the range $0 \leq \gamma^2(\omega) \leq 1$. It provides a frequency-dependent measure of how strongly the output is linearly related to the input.

The coherence is shown in Fig. 2.7. Values close to one indicate a reliable estimate, whereas lower values indicate that noise or disturbances dominate the measurement.

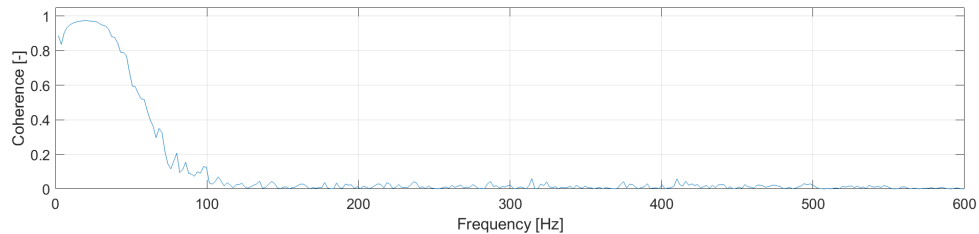


Figure 2.7: Magnitude-squared coherence between input and output signals. High values indicate a reliable transfer function estimate.

2.3 Control Loop Calculations

Although the flight controller consists of several nested control loops, their fundamental structure is identical. Each loop follows the same basic feedback control principle and differs mainly in the controlled variable and the chosen controller parameters. Therefore, instead of analysing every loop individually from the beginning, a simplified model is used to explain the underlying calculations and controller behaviour.

The simplified control structure used for the analysis consists of a controller C and a plant P , which describes the dynamic behaviour of the system being controlled.

Using this simplified model allows the general behaviour of the control loop and the calculation of the controller terms to be explained in a clear and structured way. The same principles are later applied to the individual control loops implemented in the flight controller.

2.3.1 Calculation of the Control Loop Components

After the signal processing steps, the control loop components can be calculated. As in the signal processing stage, the goal is to mitigate the influence of disturbances and noise on the calculated control loop components. Therefore, only the input signal of the control loop is used for the estimation.

The advantage of this approach is that

- the input signal is not affected by the back-calculation of the control loop and is therefore not influenced by disturbances or noise within the loop.
- the input signal is not affected by filtering steps, since it is not measured but taken directly from the software. Consequently, it is not subject to measurement noise or disturbances.

This approach allows a more precise estimation of the transfer function. However, it also means that the transfer functions of the individual control loop components cannot be calculated directly. Instead, they must be obtained through mathematical manipulations.

For example, consider a simplified control loop consisting of a controller $C(s)$ and a plant $P(s)$.

To calculate the transfer function of the controller $C(s)$, the following relation can be used

$$C(s) = \frac{Y(s)}{E(s)} = \frac{Y(s)}{R(s) - Y(s)} = \frac{Y(s)}{R(s)} \cdot \frac{1}{1 - \frac{Y(s)}{R(s)}} = \frac{G(s)}{1 - G(s)}.$$

To calculate the transfer function of the plant $P(s)$, the following relation can be used

$$P(s) = \frac{Y(s)}{U(s)} = \frac{Y(s)}{E(s)} \cdot \frac{E(s)}{U(s)} = C(s) \cdot \frac{1}{C(s)} = 1.$$

2.3.2 Closed-Loop Performance Metrics

After determining the plant transfer function, the next step is to analyze the behaviour of the overall system when a controller is applied. This is achieved by evaluating the closed-loop transfer functions, which describe the system response, disturbance sensitivity, disturbance rejection capability, and the required controller effort. These measures provide valuable insights into the stability, robustness, and performance of the controlled system.

The closed-loop transfer function from the reference input to the output is given by

$$G_{yr}(\omega) = \frac{C(\omega)P(\omega)}{1 + C(\omega)P(\omega)}.$$

The sensitivity function, which describes how sensitive the system output is to disturbances and model uncertainties, is defined as

$$S(\omega) = \frac{1}{1 + C(\omega)P(\omega)}.$$

The compliance function, representing the system response to disturbances acting on the plant, is given by

$$SP(\omega) = P(\omega) S(\omega).$$

Finally, the controller effort describes the influence of the reference input on the controller output and is defined as

$$SC(\omega) = C(\omega) S(\omega).$$

2.3.2.1 Step Response Analysis

To be added.

3 Methodology

In this chapter, the practical application of the theoretical concepts introduced in the previous chapter is described. The focus is on how these principles are implemented in the context of tuning the control loops of a Betaflight-based flight controller. The methodology is structured around the different control loops, starting with the angle controller, followed by the altitude hold controller and the position controller. For each control loop, the specific implementation details and tuning procedures are described.

3.1 Angle Mode

3.1.1 Difference between Angle and Acro Mode

The Betaflight software provides three main flight modes: Angle, Horizon, and Acro.

In Angle mode, the drone's tilt angle is directly linked to the stick position. The maximum tilt is limited to a predefined value, preventing flips and making this mode particularly suitable for beginners.

In contrast, Acro mode provides full manual control. In this mode, stick inputs define an angular rate rather than a target angle, meaning the drone continues rotating as long as the stick remains deflected. This allows aggressive maneuvers such as flips and rolls but also requires greater pilot experience.

During the project thesis, an offline tuner for the angular rate controller was further developed to systematically optimize the rate control loop and improve Acro mode performance. The tuner was subsequently extended to also support Angle mode, enabling improved stability and responsiveness while maintaining intuitive handling for beginner pilots.

3.1.2 Control Loops in Angle Mode

In Angle mode, the angular rate controller is extended by an outer angle control loop, resulting in a cascaded control structure.

The stick input initially generates a rate setpoint. Unlike in Acro mode, this rate setpoint is then converted into a target angle. This target angle serves as the reference input for the inner angular rate controller.

The cascaded architecture separates attitude control from angular rate stabilization. While the outer loop ensures convergence toward the desired orientation, the inner loop provides fast dynamic stabilization by regulating the angular velocity. As a result, the system achieves improved disturbance rejection and more intuitive handling characteristics, since the pilot directly commands an angle rather than a rotational rate.

The Angle mode controller therefore consists of two cascaded loops:

- The outer angle loop, which computes the angle error
- The inner angular rate loop, which computes the rate error

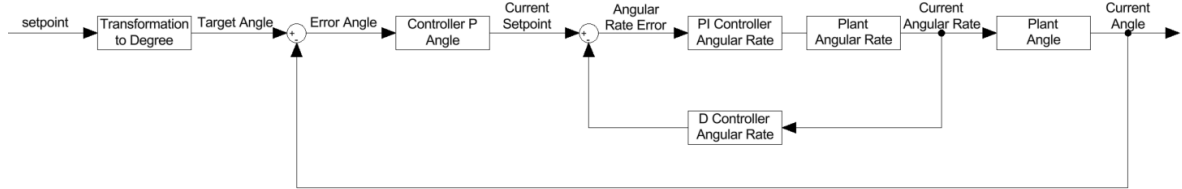


Figure 3.1: Cascaded control structure of the Betaflight Angle mode controller.

The angle control loop consists of a proportional controller with an additional feedforward term. Within the offline tuning framework, however, only the proportional controller is optimized. The feedforward path is included in the control structure for completeness but was deliberately disabled during the tuning process.

3.1.3 Derivation of the Target Angle

The target angle is obtained by scaling the commanded angular rate with respect to the maximum achievable rate and the configured maximum tilt angle

$$\theta_{\text{target}} = \theta_{\text{limit}} \cdot \frac{\omega_{\text{set}}}{\omega_{\text{max}}} \quad (0.1)$$

where

- θ_{limit} is the maximum allowed tilt angle (e.g., 60°),
- ω_{set} is the commanded angular rate derived from the stick input,
- ω_{max} is the maximum achievable angular rate determined by the rate configuration.

This relation ensures a proportional mapping between stick deflection and target angle.

For example,

$$\omega_{\text{set}} = \omega_{\text{max}} \Rightarrow \theta_{\text{target}} = \theta_{\text{limit}} \quad (0.2)$$

$$\omega_{\text{set}} = 0 \Rightarrow \theta_{\text{target}} = 0 \quad (0.3)$$

Thus, the stick position scales the target angle continuously within the range

$$-\theta_{\text{limit}} \leq \theta_{\text{target}} \leq +\theta_{\text{limit}} \quad (0.4)$$

3.1.4 Chirp Signal for System Identification

As already mentioned in the gyro control tuning section, a chirp signal is injected into the control loop.

If the chirp signal were added in the same way as for the gyro control tuning, it would not be possible to directly calculate the transfer function of the complete system from the target angle to the measured angle. The chirp signal would be added to the current setpoint and therefore inside the closed control loop. Consequently, the system would have two inputs: the stick input and the injected chirp signal.

This situation is illustrated in Figure 3.2, which shows the control loop structure currently implemented in the Betaflight firmware.

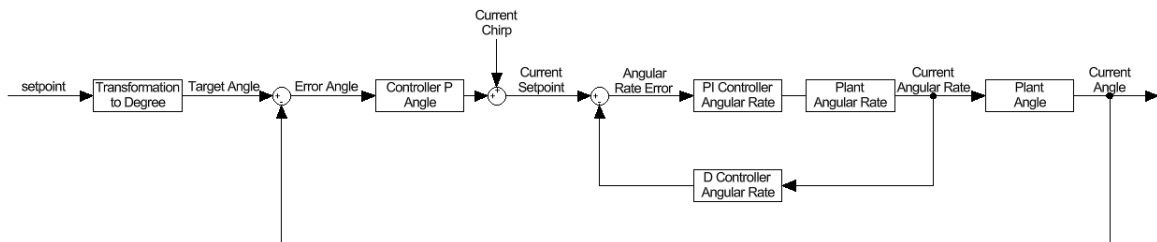


Figure 3.2: Current control loop structure in Betaflight when the chirp signal is injected inside the closed loop.

To overcome this issue, a small modification in the firmware allows the chirp signal to be added directly to the setpoint before it is converted into the target angle. Due to the transformation from the setpoint to the target angle, the parameters of the chirp signal do not need to be adjusted.

The modified control structure used for the offline tuning framework is illustrated in Figure ?? . In this configuration, the chirp signal is injected before the transformation from the rate setpoint to the target angle, allowing the transfer function from the target angle to the measured angle to be estimated directly.

Some older drones are only capable of flying in Acro mode and therefore do not support Angle mode. In this case, the chirp signal can still be added to the setpoint. Since the angle control loop and the transformation from the setpoint to the target angle are disabled in Acro mode, the resulting control loop structure is identical to the one used for the gyro control tuning.

This approach allows the same chirp signal to be used for both modes. As a result, the implementation is simplified and the tuning process becomes more straightforward, since the same excitation signal can be applied without considering structural differences in the control loop.

3.1.5 Measurement of the Control Loop Components

Based on the modified control structure, the dynamic behaviour of the Angle mode control loop can be analysed using the recorded input and output signals. Due to the chirp signal being injected before the transformation to the target angle, the system can be treated as a single-input single-output system from the target angle to the measured angle.

After applying the signal processing steps described in the previous chapter, the frequency response of the closed control loop is estimated. Using the Fourier transforms of the target angle $\Theta_{\text{target}}(\omega)$ and the measured angle $\Theta(\omega)$, the closed-loop transfer function is obtained as

$$G_{cl}(\omega) = \frac{\Theta(\omega)}{\Theta_{\text{target}}(\omega)}.$$

This transfer function describes the complete behaviour of the cascaded control structure, including both the outer angle controller and the inner angular rate controller, as well as the plant dynamics.

For a standard feedback system, the closed-loop transfer function can be expressed as

$$G_{cl}(\omega) = \frac{C(\omega)P(\omega)}{1 + C(\omega)P(\omega)},$$

where $C(\omega)$ represents the combined controller dynamics and $P(\omega)$ denotes the plant.

By rearranging this expression, the corresponding open-loop transfer function can be computed

$$C(\omega)P(\omega) = \frac{G_{cl}(\omega)}{1 - G_{cl}(\omega)}.$$

This formulation allows the combined dynamics of the controller and the plant to be determined directly from measurement data.

Since only the overall closed-loop behaviour is observed, the individual transfer functions of the controller and the plant cannot be uniquely separated without additional information. However, the open-loop transfer function fully characterizes the stability and performance of the control loop and is therefore sufficient for the subsequent tuning process.

In practice, the estimated open-loop transfer function is used to evaluate bandwidth, phase margin, and disturbance rejection properties of the system, forming the basis for the offline tuning procedure.

3.1.6 Theoretical Expectation

If we consider the fundamental relationship between angular rate and orientation, the angle corresponds to the time integral of the angular rate. Hence, the measured angle can be written as

$$\theta_{\text{measured}}(t) = \int \omega_{\text{measured}}(t) dt. \quad (3.1)$$

Assuming zero initial conditions and applying the Laplace transform, this relationship in the frequency domain becomes

$$\Theta_{\text{measured}}(s) = \frac{1}{s} \Omega_{\text{measured}}(s). \quad (3.2)$$

This expression shows that the angle is obtained by integrating the angular rate, which corresponds to a multiplication by $\frac{1}{s}$ in the Laplace domain. Consequently, from a theoretical point of view, the Angle plant behaves as an ideal integrator with transfer function

$$P_{\text{Angle}}(s) = \frac{1}{s}. \quad (3.3)$$

Declaration of Generative AI Use

The following generative AI systems were used during the preparation of this thesis. Their use was limited to the functions described below. All generated content was critically reviewed and verified by the authors.